8 **(a)** Find $\int \sin\theta \cos^n\theta \, d\theta$, where $n \neq -1$. [2]

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Let $I_{m,n} = \int_0^{\frac{1}{2}\pi} \sin^m\theta \cos^n\theta \, d\theta$.

(b) Show that, for $m \geqslant 2$ and $n \geqslant 0$,

$$I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}.$$ [5]

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(c) By considering the binomial expansion of $\left(z+\frac{1}{z}\right)^5$, where $z = \cos\theta + \mathrm{i}\sin\theta$, use de Moivre's theorem to show that

$$\cos^5\theta = a\cos 5\theta + b\cos 3\theta + c\cos\theta,$$

where a, b and c are constants to be determined. [5]

(d) Using the results given in parts **(b)** and **(c)**, find the exact value of $I_{2,5}$. [4]